

Causal Volatility Management and Cost-Aware Exposure for Stable Covariance Portfolios

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Abstract

Covariance forecasts that react quickly to new returns can induce excessive trading, while smoothing the entire forecast can delay risk reduction. We study a modular response: estimate a stable constrained portfolio direction from a smoothed covariance forecast, manage its total exposure using a current risk forecast with a causal normalization, and execute exposure changes through a one-dimensional convex problem with a turnover penalty. The covariance predictor is the published CM-IEWMA model of Johansson et al. (2023); the contribution is an implemented and tested allocation-and-execution extension, rather than a new unrestricted covariance estimator.

Using the authors' 49-industry return panel, we develop the method on 1976–1994, validate the fixed configuration on 1995–2004, and carry every forecast and account state into a final 2005–2022 evaluation. At a proportional cost of 5 basis points per side, the final-period net excess Sharpe is 0.744, compared with 0.625 for native CM-IEWMA, 0.544 for covariance smoothing and 0.484 for a same-risk equal-weight benchmark. Maximum drawdown is 7.50%, versus 14.73% for native CM-IEWMA and 12.13% for same-risk equal weight. Relative to matched direct volatility management, gross annual turnover falls from 5.97 to 2.83 times NAV, while Sharpe increases from 0.730 to 0.744. A simple exposure buffer remains a very strong control and slightly exceeds the candidate's point Sharpe. Paired block-bootstrap intervals include zero for the return improvements. The evidence supports a useful turnover–risk trade-off and a reproducible implementation, but does not establish universal or statistically decisive return superiority.

Keywords: covariance forecasting; volatility management; portfolio turnover; convex optimization; chronological evaluation.

1 Research problem and relationship to prior work

The downstream decision matters as much as the accuracy of a risk estimate. A covariance forecast may improve a likelihood score while moving constrained portfolio weights enough to lose the benefit to trading costs. The practical question in this study is therefore narrower than predicting every entry of a large covariance matrix: how can a portfolio retain a stable composition while reacting to changing aggregate risk?

The distinction is particularly relevant when the window available for adaptation is short relative to the number of assets. Our application has 49 industry portfolios, whereas the published CM mixture uses ten mature forecast–return pairs to adapt five expert weights. Long-history experts remain essential. This setting is not a claim that an unrestricted 49-by-49 covariance matrix is identified by ten observations.

Johansson et al. (2023) provide the underlying CM-IEWMA estimator and explicitly discuss smoothing covariance predictions in Chapter 9. Moreira and Muir (2017) motivate inverse-variance exposure management. Boyd et al. (2017) describe convex trading objectives that balance risk and transaction costs. We build on these ideas and make three implementation choices explicit: separate the slow allocation direction from the fast exposure decision; normalize inverse-variance exposure using only previously issued risk forecasts; and optimize the actual exposure coordinate instead of interpolating an entire constrained target vector. None of covariance smoothing, inverse-variance scaling, or an L1 trading penalty is claimed as a first invention.

Our empirical contribution is a matched comparison against the original risk models, strong smoothing controls, direct and buffered exposure rules, and a separately developed scalar-risk estimator. All methods share a return panel, calendar, cash convention, constraints and exact fee-accounting implementation. The original Chapter 9 experiment uses Fama–French factor returns; its smoothing method is transferred here

to the author’s 49-industry panel. We do not label this transfer as a numerical replication of the Chapter 9 table.

2 Information, data and benchmark portfolio

2.1 Calendar and information set

Let $r_t \in \mathbb{R}^{49}$ denote the return vector observed by the end of day t . A decision issued at t is held over the next observation, $t + 1$. Every covariance, smoothing update and exposure target is computed from information available through t . The return r_{t+1} is consumed only after the target has been formed.

We use the decimal daily return files distributed with the authors’ public code, pinned to commit 8892f0311f2e047ecf465f0995540bee13e5de7d. The panel comprises 49 US industry portfolios, not 49 directly tradable individual securities. Risk-free returns come from the accompanying factor file. The evaluation ends on 30 December 2022; later rows in the downloaded source are excluded.

Stage	Holding dates	Days	Role
Development	1976-01-02–1994-12-30	4,803	Method development and fixed H selection
Validation	1995-01-03–2004-12-31	2,519	Fixed-parameter chronological comparison
Final	2005-01-03–2022-12-30	4,531	Final comparison without retuning

All three stages use daily sequential forecasts. Forecast states, covariance smoothing, normalizers, actual holdings and cash are carried across the boundaries. We do not restart an account with cash at the beginning of validation or final evaluation. Stage-specific drawdowns and compound growth are reported after normalizing the stage’s initial wealth to one.

Development included several unsuccessful alternatives. Those screens are not relabeled as successes. The candidate configuration was fixed before the final period was evaluated, but the data are public and the original paper’s full-period benchmark tables had already been consulted. Thus the chronological evaluation does not constitute a globally unseen research holdout or a live trading record.

2.2 Published covariance inputs

The original CM-IEWMA predictor combines five IEWMA experts with volatility/correlation half-lives (21, 63), (63, 125), (125, 250), (250, 500) and (500, 1000) days. Following the source implementation, the fastest expert receives a 5% diagonal addition. Expert covariance estimates use their own volatility scales.

For an expert covariance $\Sigma_{t,k}$, let $L_{t,k}$ be the lower Cholesky factor of its precision matrix. For mixture weights a in the five-dimensional simplex,

$$L_t(a) = \sum_{k=1}^5 a_k L_{t,k}, \quad \widehat{\Sigma}_t = \{L_t(a_t) L_t(a_t)^\top\}^{-1}.$$

The weights minimize the zero-mean Gaussian negative log-likelihood over ten **matured, previously issued** expert forecasts. The learning objective is convex in a . A return is scored against the expert factors that existed before that return, not against a factor fitted after observing it. A single IEWMA model with half-lives (125, 250) is retained as a secondary input. We leave these underlying estimation objectives unchanged.

2.3 Original allocation and cash convention

Given a covariance S , the common base portfolio solves

$$b(S) = \arg \min_b b^\top S b \quad \text{subject to} \quad \mathbf{1}^\top b = 1, \quad \|b\|_1 \leq 1.6, \quad -0.10 \leq b_i \leq 0.15.$$

The native benchmark scales the resulting direction to 5% predicted annual volatility. Risky weights are θb , and cash is $1 - \theta$. The base constraints apply to b , not directly to the final leveraged risky vector. We

follow the original uncapped exposure convention, allowing negative cash. This does not model a broker's margin requirement or stock-borrow availability.

3 Stable allocation and causal exposure control

3.1 Separate portfolio direction from current risk

We smooth the issued covariance forecast with a 250-day half-life:

$$\bar{\Sigma}_t = \beta_b \bar{\Sigma}_{t-1} + (1 - \beta_b) \hat{\Sigma}_t, \quad \beta_b = 2^{-1/250}, \quad b_t = b(\bar{\Sigma}_t).$$

The first development forecast initializes this post-processing filter; the filter is then carried continuously. Long-history native estimators are already initialized from the preceding source history. The current risk of the stable direction is evaluated with the *unsmoothed* forecast,

$$v_t = b_t^\top \hat{\Sigma}_t b_t > 0.$$

This is a two-timescale design: the portfolio direction is stable, but its risk estimate can respond without waiting for the 250-day smoothing filter. It differs from scaling b_t using $b_t^\top \bar{\Sigma}_t b_t$.

3.2 Causal normalization of inverse-variance exposure

Set $s = 0.05/\sqrt{252}$. For $p \in \{1/2, 1\}$, form an exposure target

$$\theta_t^* = \frac{s v_t^{-p}}{\sqrt{M_{t-1,p}}}, \quad M_{t-1,p} = \frac{\sum_{u < t} \beta_m^{t-1-u} v_u^{1-2p}}{\sum_{u < t} \beta_m^{t-1-u}}, \quad \beta_m = 2^{-1/252}.$$

At the first development observation, before any earlier issued risk is available, initialize M with the current known v_0^{1-2p} . Subsequently the current v_t is added to the normalizer only after the current target has been determined.

For $p = 1/2$, the normalizer is one and the target is conventional constant predicted-volatility scaling. For $p = 1$, exposure is inverse variance with a lagged scale normalization. The normalization avoids using a full-sample realized volatility constant. It does **not** make predicted risk constant each day: $\theta_t^* v_t = s^2 v_t^{-1} / M_{t-1,1}$ for $p = 1$. Nor does it guarantee future realized volatility or returns. This is our causal implementation of volatility management, not an exact numerical replication of Moreira and Muir's portfolio construction.

3.3 Trading-cost-aware exposure adjustment

Let $h_t \in \mathbb{R}^{49}$ be the risky holdings immediately before rebalancing, expressed as fractions of current pre-fee NAV. For each physical cost scenario $c \in \{0, 0.0005, 0.001\}$, solve

$$\min_{\theta \in [0.9\theta_t^*, 1.1\theta_t^*]} \text{quad} F_t(\theta) = \frac{\gamma H v_t}{2} (\theta - \theta_t^*)^2 + c \sum_{i=1}^{49} |\theta b_{t,i} - h_{t,i}|, \\ \gamma = 10.$$

The first term penalizes displacement from the desired exposure; the second discourages costly trades. H is a tracking-strength scale, not a realized holding period. The development grid was $H \in \{20, 252, 1260\}$. The primary $p = 1$ CM configuration selected $H = 20$ by 5-bp development certainty equivalent and kept it throughout later stages. The band is a relative exposure band. In particular, it is not a fixed 4.5%–5.5% volatility band when $p = 1$.

The objective is a local tracking-and-trading model. It does not estimate expected returns, globally solve an unconstrained dynamic investment problem, or optimize the exact nonlinear fee-account equation. The latter is used to evaluate every executed target consistently.

3.4 Scalar solution and certificate

For nonzero b_i , define the breakpoints $z_i = h_i/b_i$ and masses $k_i = c|b_i|$. Up to terms independent of θ ,

$$F_t(\theta) = \frac{1}{2}A(\theta - \theta_t^*)^2 + \sum_i k_i |\theta - z_i|, \quad A = \gamma H v_t > 0.$$

On an interval between sorted breakpoints, the derivative is linear and its root is

$$\theta = \theta_t^* - \frac{\sum_{z_i < \theta} k_i - \sum_{z_i > \theta} k_i}{A}.$$

Proposition. The bounded scalar problem has a unique minimizer. Its solution is obtained by finding the interval or breakpoint where the monotone one-sided derivatives bracket zero, with the appropriate endpoint condition when the optimum is at a bound. Sorting costs $O(N \log N)$; a cumulative scan then costs $O(N)$.

Proof. The quadratic term is strictly convex because $A > 0$, and the absolute-value terms are convex. The sum is therefore strictly convex on a nonempty compact interval and has a unique minimizer. Its subdifferential is monotone. Within a smooth interval, setting the derivative to zero gives the displayed root; at a breakpoint the left and right derivatives describe the full subgradient interval. The lower endpoint is optimal when its right derivative is nonnegative, and the upper endpoint is optimal when its left derivative is nonpositive. These cases exhaust a monotone subdifferential. Each breakpoint and cumulative mass is visited at most once after sorting. This proves the proposition.

For an admissible subgradient g , a computable global lower bound on the *working objective* is

$$F_t(\theta) - G_t, \quad G_t = g \left(\theta - \arg \min_{x \in [l, u]} gx \right) \geq 0.$$

This certificate concerns the scalar problem at the fixed direction. It does not certify an optimal asset allocation over all risky directions. The development implementation initially encountered a floating-point tie between adjacent breakpoints; selecting by monotone derivatives corrected it without changing financial parameters or the 10^{-12} gap threshold. Twenty numerical checks, including 576 independent scalar references, passed. Later daily archives retain the solver gap; they do not retain every development diagnostic field.

3.5 Why changing the decision coordinate matters

A preliminary alternative interpolated an entire target vector: $w(a) = (1 - a)h + a\tilde{w}$. If the drifted portfolio violates a normalized upper bound by $\delta > 0$ and the target lies exactly on that bound, the violation along the ray equals $(1 - a)\delta$. Every $a < 1$ is infeasible, forcing a full trade even when the unconstrained cost optimum calls for little trading.

Optimizing the scalar exposure along a freshly feasible base direction avoids this particular geometry: for every positive θ , normalized risky weights remain b_t . It does not make the final leveraged exposure independently satisfy the unscaled base bounds. The empirical contribution of this redesign is assessed against a simple clipped-exposure buffer, since a mathematical optimizer should not receive credit for benefits available from an inexpensive rule.

4 Exact account and experimental controls

4.1 Fees, cash and drift

Write $w_t = (\theta_t b_t, 1 - \theta_t)$ for the full target including cash, and let h_t^{full} be the drifted pre-trade portfolio. If q_t is post-fee NAV divided by pre-fee NAV, the accounting engine solves

$$q_t + c \sum_{i=1}^{49} |q_t w_{t,i} - h_{t,i}| = 1.$$

The risky turnover charged to the account is the sum inside this equation, not the optimizer’s approximate $\sum |\theta b - h|$. Cash has zero trading cost. Positive and negative cash earn or pay the signal-date risk-free rate from the original code. For the next return vector R_{t+1} including that cash return,

$$V_{t+1} = V_t q_t (1 + w_t^\top R_{t+1}), \quad h_{t+1}^{\text{full}} = \frac{w_t \odot (1 + R_{t+1})}{1 + w_t^\top R_{t+1}}.$$

Each cost rate has a separate account and, for the optimizer and buffer, a separate exposure path because holdings affect the next decision. This comparison excludes market impact, financing spreads, borrow fees and the constituent trades inside industry portfolios. The results are historical research simulations, not measured implementable net performance at a stated fund capacity.

4.2 Controls and the parallel estimation route

The exposure experiment contains 32 prespecified identities: two covariance inputs, two powers, and eight execution choices. The latter are three values of H , direct execution, three clipped exposure EMAs with half-lives 5/21/63, and a buffer that clips current net risky exposure to the same relative band. All $p = 1/2$ controls use the same base direction. We also retain native CM and IEWMA, Chapter 9 covariance smoothing, and target-weight EMA-250.

A parallel route estimates only the scale error of a slowly changing portfolio. For a direction and variance issued at u , it observes the mature value $y_u = b_u^\top r_{u+1}$ and $z_u = y_u^2 / v_u$. With decayed sums W, Z, Y over these mature values, matched estimators use $Y/W, v_t Z/W$, or $v_t(10+Z)/(10+W)$. The last expression is a fixed inverse-Gamma working-prior shrinkage estimate, with prior shape 6 and scale 5. Half-lives 10/21/63 were developed before validation; H10 is carried as the selected alternative. The likelihood is a working approximation, not a calibrated dynamic Bayesian model under financial dependence.

The combined comparison reports 60 identities and three cost scenarios in each chronological stage. These are dependent experimental configurations, not 60 independent replications. Additional exploratory failures remain in compact research records and are not used to change the meaning of earlier unsuccessful screens.

4.3 Performance definitions

Let $x_t = r_t^{\text{net}} - r_{f,t}^{\text{signal}}$. We report annual arithmetic excess return $252\bar{x}$, excess Sharpe $\sqrt{252}\bar{x}/s_x$, net certainty equivalent $252(\bar{x} - 5s_x^2)$, total-return CAGR, realized total-return volatility, maximum drawdown and annual gross risky turnover. The standard deviations use sample normalization. CAGR includes the cash return and differs from annual arithmetic excess return. All tables below use 5 bp per side unless stated otherwise.

5 Empirical results

5.1 Chronological comparison

Stage	Cost-aware H20 Sharpe	Direct $p = 1$ Sharpe	Native CM Sharpe	Covariance EMA-250 Sharpe	Cost-aware turnover	Direct turnover	Cost-aware drawdown
Development, 1976–1994	0.857	0.815	0.431	0.753	3.15	8.14	11.47%
Validation, 1995–2004	0.699	0.686	0.391	0.462	2.91	6.96	14.18%
Final, 2005–2022	0.744	0.730	0.625	0.544	2.83	5.97	7.50%

The candidate maintains positive point improvements over direct execution across all three chronological stages. Its final-period total-return CAGR is 4.79%, annual arithmetic excess return is 3.64%, realized volatility is 4.89%, and maximum drawdown is 7.50%. These are the outcomes of a low-risk allocation experiment, not a high-return stock-selection strategy.

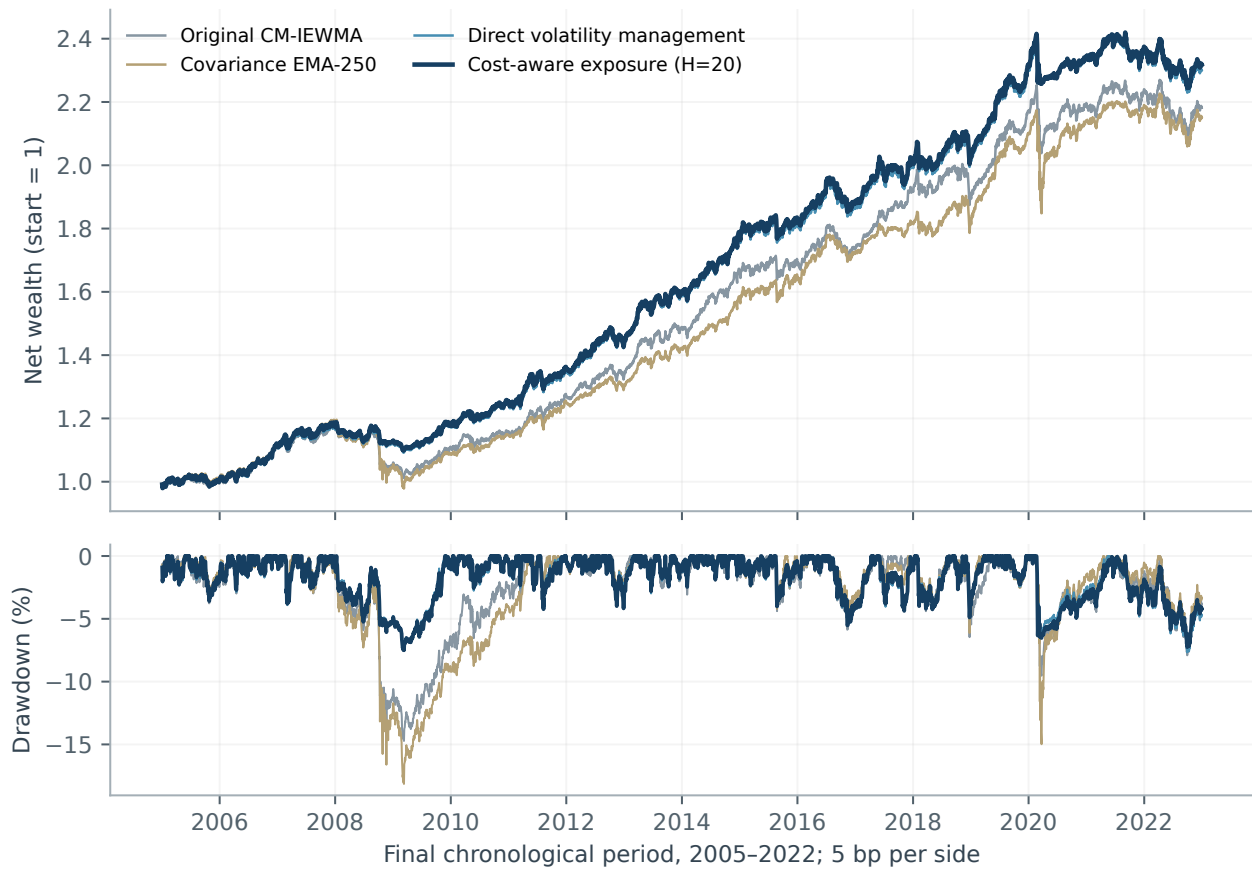


Figure 1: Net wealth and drawdown in the final period. Every series uses the same signal-date cash convention and 5-bp proportional risky trading cost.

5.2 Stronger controls and attribution

Method, final 2005–2022	Net CE	Excess Sharpe	CAGR	Volatility	Max drawdown	Annual turnover
Native	1.91%	0.625	4.44%	5.32%	14.73%	20.20
CM-IEWMA						
Covariance	1.46%	0.544	4.35%	6.04%	18.15%	1.60
EMA-250						
Direct	2.38%	0.730	4.74%	4.93%	7.68%	5.97
inverse-variance						
exposure						
Cost-aware	2.44%	0.744	4.79%	4.89%	7.50%	2.83
exposure, H20						
Simple exposure	2.45%	0.746	4.80%	4.89%	7.45%	3.00
buffer						
Exposure EMA-5	2.45%	0.743	4.83%	4.95%	7.66%	3.49
Same-risk equal	1.17%	0.484	3.51%	5.00%	12.13%	1.56
weight						

Separating the risk scale from the smoothed direction accounts for much of the gain over plain covariance smoothing. Inverse-variance management reduces the final drawdown further relative to the constant predicted-volatility control. Adding the cost objective to matched direct $p = 1$ execution reduces turnover by 52.57%, raises Sharpe by 0.013 and adds 5.65 bp to annual certainty equivalent. The incremental benefit is modest; it should not be conflated with the larger gain of the entire architecture over native CM. The final-period risky exposure is not continuously fully invested: its mean is 47.26%, median 46.87%, 5th–95th percentile range 15.43%–78.71%, and maximum 115.62%; exposure exceeds 100% on 0.42% of days. Cash earns the signal-date risk-free rate, and negative cash follows the original uncapped convention.

Equal weight is compared on two distinct bases. The primary equal-weight benchmark is dynamically scaled by the same current CM risk forecast to approximately 5% realized volatility, so its Sharpe 0.484 and drawdown 12.13% are comparable to the candidate’s 0.744 and 7.50%. A daily rebalanced, fully invested equal-weight portfolio has 9.45% CAGR but 20.82% volatility, 56.70% drawdown and Sharpe 0.482;

its higher raw return comes with roughly four times the candidate's risk. It is therefore a scale reference, not evidence that either method dominates in dollar return.

The buffer is especially informative. Its final point Sharpe and certainty equivalent are slightly higher than the candidate's. Exposure EMA-5 has a slightly higher certainty equivalent as well. The candidate has somewhat lower turnover than these two controls, but does not dominate them. This comparison supports cost efficiency as a useful engineering result while limiting the claim of a superior optimizer.

5.3 Costs, parameter sensitivity and secondary input

Method	Metric	0 bp	5 bp	10 bp
Cost-aware H20	Sharpe	0.791	0.744	0.715
	Net CE	2.68%	2.44%	2.30%
	Turnover	5.97	2.83	2.83
Direct $p = 1$	Sharpe	0.791	0.730	0.670
	Net CE	2.68%	2.38%	2.09%
	Turnover	5.97	5.97	5.97
Simple buffer	Sharpe	0.777	0.746	0.715
	Net CE	2.60%	2.45%	2.30%
	Turnover	3.00	3.00	3.00

The principal $H = 20$ remains fixed. The full archive includes $H = 252$ and $H = 1260$, both powers, all exposure EMAs, both input models and all fee accounts. The selected CM $p = 1$ method has similar final-period Sharpe across the three tracking strengths, but turnover changes materially. We therefore describe local outcome stability without claiming universal parameter insensitivity.

The identical H20 configuration with single IEWMA yields final Sharpe 0.593 at 5 bp. It does not replicate the CM result, and it does not outperform native CM. Transferability of the architecture depends on the underlying risk input. The parallel scalar-shrinkage method also has a mixed result: it reduces turnover relative to raw scalar variance estimation and becomes preferable in certainty equivalent at 10 bp, but it does not beat that matched control at 5 bp.

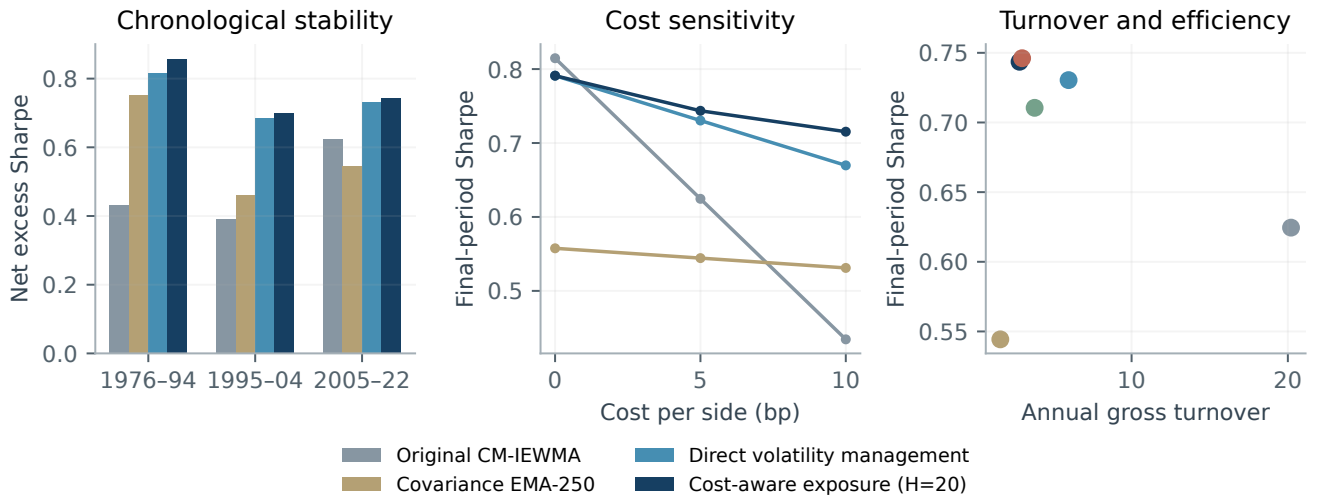


Figure 2: Chronological Sharpe, cost sensitivity and final-period turnover. The cost-aware and buffered methods have similar risk-adjusted outcomes, which limits an optimizer-specific superiority claim.

5.4 Uncertainty

We use 2,999 paired circular moving-block bootstrap replicates on the final-period daily net excess-return vectors. The same sampled indices apply to every compared method, preserving their contemporaneous dependence. We report block lengths 21, 63 and 126 days; these choices assess the sensitivity of approximate intervals to serial dependence and do not change the candidate or its parameters.

Control	Δ Sharpe	95% interval	Δ annual CE	95% interval
Native CM-IEWMA	0.119	[-0.070, 0.306]	0.534 pp	[-0.418, 1.574] pp

Control	Δ Sharpe	95% interval	Δ annual CE	95% interval
Covariance EMA-250	0.199	[-0.071, 0.418]	0.977 pp	[-0.406, 2.717] pp
Direct inverse-variance exposure	0.013	[-0.020, 0.047]	0.056 pp	[-0.103, 0.214] pp
Simple exposure buffer	-0.003	[-0.009, 0.004]	-0.011 pp	[-0.045, 0.021] pp

The 63-day intervals include zero for improvements against native CM, covariance smoothing, direct execution and the buffer. The study therefore reports historical point improvements and turnover reductions, not statistically established alpha or universal dominance. A public benchmark panel, many prior development attempts, dependence in returns and only one final market history further limit inference. A new dataset or a genuinely forward paper account is the appropriate next test.

6 Implementation and reproducibility

All research artifacts are under the user’s CV workspace; the existing ETF and futures projects were not modified or rerun. The research code uses Python, NumPy, SciPy, CVXPY and fixed-objective numerical routines. The public author data, frozen parameter records, source hashes, full tables, block checkpoints and plotting code are retained with the final manuscript.

The continuous run performs four constrained GMV solves per day for the two native and two smoothed directions. All exposure methods reuse those forecasts. Later covariance and scalar-account evaluation completes 28 identities; the separate exposure evaluator completes 32. Independent review recomputed 360 validation/final metric rows exactly from saved paths and checked the two state seams. Scalar maturity counts carry from 4,803 to 7,322 to 11,853; native states and actual holdings carry without a reset. Accounting discrepancies were at floating-point precision. This verifies the implemented chronological experiment, not the realism of every trading assumption.

A minimal replay sequence is to run the fixed development scripts, the shared chronological continuation for validation and final, and then the corresponding exposure continuation. The archive’s `REPRODUCE.md` specifies paths and ordering; completed output directories are protected against silent overwrite. The final tables and figures are regenerated by `analyze_final.py`. The same-risk and fully invested equal-weight checks are stored separately in `equal_weight_benchmark.json`.

7 Conclusions and next steps

A stable allocation direction with faster scalar risk management addresses a practical conflict between risk responsiveness and transaction costs. On the author-supplied industry panel, the fixed configuration improves the historical Sharpe and drawdown of the native and covariance-smoothed benchmarks. Its one-dimensional convex execution substantially reduces turnover relative to direct exposure targeting while preserving a similar risk profile.

The strongest simple buffer remains competitive. Accordingly, the defensible result is an explicit, reproducible portfolio-control design with a useful measured trade-off, rather than a claim that a new optimizer universally beats all alternatives. Extending the method to alpha-driven stock selection requires point-in-time universes, realistic execution, sector and factor exposure constraints, stock-borrow and financing assumptions, and a separately validated expected-return model. That extension has not been executed in this study.

Research status, responsibility and availability

This document is an internal research-portfolio manuscript. It has not been peer reviewed or submitted to a journal. AI-based coding agents assisted literature discovery, implementation, experiment orchestration, consistency checks and language editing. The named researcher directed the project and remains responsible for checking the analysis, deciding whether to circulate it, and making any authorship, funding, conflict-of-interest or venue-specific declarations. The manuscript must not be represented as an accepted publication.

The underlying 49-industry and factor-return files are distributed with the cited authors’ code. The extension code, frozen protocols, source hashes, result tables and independent computational review are retained in the local research archive. No public release of the extension archive is asserted. The paper contains no participant data, clinical data or ethics-governed experiment.

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